

IILM UNIVERSITY

School of Computer Science and Engineering



SUMMER INTERNSHIP AT DEFENCE RESEARCH AND DEVELOPMENT ORGANISATION(DRDO)

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Program- B.Tech CSE

Batch- 2024-2028

Organization Profile – Defence Research and Development Organisation (DRDO)

- India's premier organisation for defence research and development.
- Works across various scientific and engineering disciplines.
- Major areas include Artificial Intelligence, Cybersecurity, Electronics, Communications, Computer Science and other advanced technologies.
- Cybersecurity plays an important role in protecting digital systems and information.
- The internship was undertaken in the **Cybersecurity domain**.

Internship Overview

- **Internship Role:** Intern
- **Domain:** Cybersecurity
- **Organisation:** DRDO
- **Duration:** 10 June 2026 – 7 August 2026
- **Key Areas of Exposure :**
 - Artificial Intelligence
 - Large Language Models (LLMs)
 - AI Agents
 - Knowledge Retrieval
 - Python
 - SQL
 - C++
 - Virtual Machines
 - OpenCode

Objective of Internship

- Understand Artificial Intelligence and Large Language Models.
- Study the concept of AI agents.
- Understand external knowledge bases and information retrieval.
- Explore LLM-based question answering.
- Understand how relevant information can be provided to an LLM as context.
- Gain practical exposure to Python, SQL, C++, Virtual Machines and OpenCode.
- Improve programming, debugging and problem-solving skills.
- Understand security and confidentiality in a cybersecurity environment.

Technical Stack & Tools

Technology	Purpose
Python	Programming and AI-related development
LLMs	Natural-language processing and response generation
AI Agents	Intelligent task-oriented workflows
SQL	Database management and querying
C++	Programming and systems-oriented concepts
Virtual Machines	Isolated testing and experimentation
OpenCode	AI-assisted development workflow

Project Overview

Development of an LLM-Based Intelligent Knowledge Retrieval System

The project focuses on combining a **Large Language Model** with an **external knowledge base**.

Main Concept

User Query → Information Retrieval → Relevant Context → LLM → Response

Main Goal

To enable an LLM-based system to retrieve relevant information from a predefined knowledge source and use that information to generate meaningful natural-language responses.

METHODOLOGY

Project Methodology

1. **Problem Understanding**
Identify the need for an external knowledge source.
2. **Knowledge Source Preparation**
Organize relevant information into a knowledge base.
3. **Query Input**
Accept natural-language questions from the user.
4. **Query Processing**
Process the query to determine the required information.
5. **Information Retrieval**
Identify relevant information from the knowledge source.
6. **Context Preparation**
Provide the retrieved information as context to the LLM.
7. **LLM Processing & Response Generation**
Generate a natural-language response.
8. **Response Evaluation**
Evaluate relevance, clarity and consistency.

LEARNING & CONTRIBUTION

- Studied the working concepts of LLMs and AI agents.
- Explored knowledge-based information retrieval.
- Worked with Python and supporting technologies.
- Gained practical understanding of SQL and database concepts.
- Worked with Virtual Machines and isolated environments.
- Explored AI-assisted development using OpenCode.
- Improved debugging and problem-solving skills.
- Developed an understanding of responsible handling of information in a cybersecurity environment.

Challenges & Learning Outcomes

- **Challenges Faced**

- Understanding unfamiliar AI and LLM concepts.
- Understanding the interaction between knowledge sources and LLMs.
- Debugging technical issues during development.
- Working with multiple technologies and development environments.
- Understanding information retrieval workflows.

- **Learning Outcomes**

- Better understanding of AI and LLMs.
- Knowledge of AI agents and retrieval systems.
- Improved programming and debugging skills.
- Exposure to real-world technical workflows.
- Better understanding of cybersecurity and data confidentiality.

Conclusion:

The internship at DRDO provided valuable practical exposure to **Artificial Intelligence, Large Language Models, AI agents and knowledge retrieval systems**.

The project helped in understanding how an external knowledge source can be integrated with an LLM to support more relevant and context-based responses.

The internship also strengthened my **programming, debugging, problem-solving and technical skills** while providing an understanding of working in a cybersecurity-oriented environment.

THANK YOU

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